

Hotel Recommendation System

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Abstract—This document presents the design, implementation and evaluation of a content-based hotel recommendation system that exploits natural language processing (NLP) techniques on user-generated reviews. A lightweight NLP pipeline tokenizes and lemmatizes both query descriptions and hotel “tags” extracted from aggregated customer reviews. A similarity score, defined as the cardinality of the intersection between the query token set and each hotel’s tag set, ranks candidate hotels. The system returns the top ten recommendations, each accompanied by their average review score and address. Empirical evaluation on a dataset of 515,000 reviews from 1,493 European hotels demonstrates that simple set-based similarity yields meaningful, interpretable recommendations with low computational overhead.

I. INTRODUCTION

The proliferation of online hotel reviews has introduced an information overload problem for travellers seeking accommodations tailored to their preferences. Content-based recommendation, which relies on item metadata and user profiles, offers an interpretable alternative to collaborative approaches when explicit feedback or user history is sparse. This work introduces a recommendation system that ingests a natural-language description of travel requirements and outputs hotels whose aggregated review “tags” most closely match that description. The system leverages the Natural Language Toolkit (NLTK) for fundamental NLP tasks and relies on a simple set-matching algorithm for ranking.

II. METHODOLOGY

A. Dataset and preprocessing

A publicly available dataset containing 515,000 English-language customer reviews over 1,493 luxury hotels in Europe served as the data source [1]. Each record includes hotel address, average score, reviewer tags (e.g. “friendly staff”, “near metro”), and auxiliary fields. Preprocessing comprised the following steps:

- 1) Address normalization by replacing “United Kingdom” with “UK” and lowercasing all country names.
- 2) Representing all the tags as a list.
- 3) Removal of irrelevant columns, retaining only hotel name, address, average score and tag list.

B. Recommender function

Formally, let H be the set of hotels and for each hotel $h \in H$ let $T_h = \{t_1, \dots, t_k\}$ denote its tag set after tokenization, stop-word removal and lemmatization.

Given a user’s location L and free-text description Q , the recommendation function proceeds as follows. First, Q is lower-cased, tokenized into words, pruned of

English stop-words and lemmatized to form the query set D . Denote by $H_L \subseteq H$ the subset of hotels whose country field matches L (case-insensitive). For each $h \in H_L$, compute the similarity score

$$S(h, Q) = |D \cap T_h|.$$

Hotels are then sorted in descending order of $S(h, Q)$, with ties broken arbitrarily. The top ten entries are returned in a tabular form showing the hotel name, average score, address, similarity index S , and the matched token set $D \cap T_h$.

III. RESULTS AND DISCUSSION

The system was evaluated through interactive queries reflecting typical travel scenarios. For instance, a description “leisure trip for 3 days with family” in Italy yields hotels whose user tags include leisure and family (Table I). The average similarity index across the top ten recommendations ranged from 1 to 4 matched tokens, indicating selective filtering. Qualitative inspection confirmed that higher $S(h, Q)$ corresponded to hotels with reviews mentioning the desired features.

Limitations include the inability to incorporate numeric constraints (e.g., budget) and sensitivity to vocabulary mismatches; future work may explore word-embedding-based similarity or hybridizing with collaborative filtering.

IV. CONCLUSIONS AND FUTURE WORK

A content-based hotel recommender leveraging simple NLP preprocessing and set-intersection similarity provides interpretable, efficient recommendations from large review corpora. The method achieves low latency and requires minimal training or parameter tuning. Observed shortcomings in handling synonyms and budget constraints motivate extensions to distributed word representations and additional metadata integration.

V. RELATED WORK

Ebadi & Krzyzak [2] introduced a multi-aspect hotel recommender combining explicit user ratings with topic-model-based sentiment analysis to extract implicit features, achieving high accuracy at the cost of greater model complexity. Ramzan *et al.* [3] proposed an opinion-based sentiment analysis pipeline over heterogeneous hotel data, constructing a feature matrix via lexical, syntactic and semantic analyses to enhance collaborative filtering performance. Melese [4] developed a hybrid restaurant recommendation system blending content- and

TABLE I
hotel_recommender("italy", "leisure trip for 3 days with family")

	Hotel_Name	Average_Score	Hotel_Address	Similarity Index	matched
0	Hotel Da Vinci	8.1	Via Senigallia 6 20161 Milan Italy	4	{family, leisure, trip, 3}
1	UNA Hotel Century	8.6	Via Fabio Filzi 25 B Central Station 20124 Mil...	4	{family, leisure, trip, 3}
2	Rosa Grand Milano Starhotels Collezione	8.6	Piazza Fontana 3 Milan City Center 20122 Milan...	4	{family, leisure, trip, 3}
3	TownHouse Galleria	8.3	Via Silvio Pellico 8 Milan City Center 20121 M...	4	{family, leisure, trip, 3}
4	TownHouse Duomo	8.4	Via Silvio Pellico 2 Milan City Center 20121 M...	4	{family, leisure, trip, 3}
5	Carlyle Brera Hotel	8.2	Corso Garibaldi 84 Milan City Center 20121 Mil...	4	{family, leisure, trip, 3}
6	Hotel Spadari Al Duomo	9.3	Via Spadari 11 Milan City Center 20123 Milan I...	4	{family, leisure, trip, 3}
7	TownHouse 12	8.1	Piazza Gerusalemme 12 Sempione 20154 Milan Italy	4	{family, leisure, trip, 3}
8	Starhotels Ritz	8.1	Via Spallanzani 40 Central Station 20129 Milan...	4	{family, leisure, trip, 3}
9	Hotel Galileo	7.9	Corso Europa 9 Milan City Center 20122 Milan I...	4	{family, leisure, trip, 3}

collaborative-filtering, reporting an 85% success rate with random forest models. These approaches contrast with the present work's focus on minimal preprocessing and transparent similarity metrics.

REFERENCES

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